Highlights

Operating a gas-lifted well under severe slugging: time-averaged signatures from a two-fluid simulator, physics-informed surrogates of the choke and gas-lift decisions, and their operational optimisation

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- Gas-lift branch simulated with the two-fluid model, controlled by the platform gas pressure
- Sobol' campaigns from one base state with adaptive horizon: 291 runs over choke and lift gas in three days
- Time-averaged markers at eight points replace trajectories as the training data
- Physics of the averages and of the maxima cuts the residuals three-fold, the gas-rate error with few runs by two, and the overfitting
- Optimum of revenue minus compression on the surrogate, with capacity constraints on predicted maxima

Operating a gas-lifted well under severe slugging: time-averaged signatures from a two-fluid simulator, physics-informed surrogates of the choke and gas-lift decisions, and their operational optimisation

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ABSTRACT

A gas-lifted satellite well with a flowline and a riser can produce in severe slugging, in a pressure cycle or in a steady state depending on two operating decisions, the opening of the topside choke and the pressure of the lift gas at the platform, and the time-averaged production of an unstable operating point is not what a steady-state model predicts for it. This paper builds the data and the surrogate models of those two decisions and uses them to optimise the operation of the well. The well, flowline, riser and gas-lift branch are simulated with the four-equation two-fluid model of a companion study, with the flow pattern identified dynamically and the gas-lift line, annulus and check valve simulated with the same model and controlled only by the platform gas pressure. Three sampling campaigns over the two decisions, in rounds of Sobol' points every one of which starts from the final state of the base run, with an adaptive horizon that detects the established regime and captures three cycles, produced 291 settled runs in three days on a laptop over 20 to 260 bar of lift gas, plus 29 budget-limited runs in the corner of high opening and high gas pressure, where the liquid outflow at the riser top cycles every minute and the integrator crawls; each run is reduced to time-averaged markers (mean, minimum, maximum and the two extreme frequencies) of pressure, void fraction and phase flow rates at eight characteristic points of the network. The map of the decisions shows a stability boundary that runs from 80 bar of lift gas with the choke nearly closed to 125 bar with it open, continuous injection modulated by the cycle below it, and a production gain of 34 % above it at 160 bar and 40 % at 250 bar. Two surrogates of the mean liquid, gas and lift-gas rates and of their spans were trained on the markers: a neural network and a physics-informed network whose physics are the time-averaged mass balances of the well, the gas-lift branch and the valve, exact for the averages and evaluated at collocation points without simulation. With 204 training runs and 4000 epochs, the physics-informed network predicts the mean liquid rate with a test coefficient of determination of 0.96 against 0.90 for the data-only network, both predict the gas and lift-gas rates with 0.997 to 0.998, and the physics act as a regulariser: the data-only network overfits beyond a few hundred epochs, its maximum liquid rate falling from 0.79 to 0.68, while the physics-informed one, with four inequalities that tie the maxima to the means and spans, keeps that maximum at 0.82 and the liquid span at 0.76, reduces the residual of the well balance four times and halves the error of the gas rates with ten training runs; the spans of the gas rates, the signature of slugging, remain the quantity that neither network predicts reliably. The surrogate then enters an operational optimisation of the two decisions: the revenue of the mean liquid rate minus the cost of compressing the lift gas from 5 bar to the platform pressure, with the predicted maxima of the lift-gas and liquid rates bounded by the compressor and separator capacities. With a compressor of 1.75 $\mathrm{kg\,s^{-1}}$ as the capacity constraint, at the prices of the example, the compression costs one per cent of the revenue and the optimum is at about 227 bar with the compressor as the active constraint and the choke opening indifferent above 0.7; the objective is nearly flat along the constraint, smaller compressors move the optimum down the valve law to the edge of the stable region, and an operating curve, the optimal opening and margin for each platform pressure, shows the choke open from 110 to 226 bar and throttled beyond the compressor limit. The campaign is the whole cost of the method, 75 process-hours of simulation against a second of training and a fraction of a second for every optimisation, which on the simulator would take a year.

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1. Introduction

Continuous gas lift is the artificial-lift method of most deepwater satellite wells in Brazil, and its optimisation — how much gas to inject in each well of a field that shares a compressor — has a large literature built on steady-state performance curves (Souza, 2010). A well with a flowline and a riser, however, does not always have a steady state to be optimised: with the choke open and little gas it produces in severe slugging (Schmidt, Brill and Beggs, 1980; Taitel, 1986; Jansen, Shoham and Taitel, 1996), and the time-averaged production, the liquid surges that reach the separator and the pressure swings at the wellhead are quantities of the cycle, not of the unstable steady state that a nodal model would return for the same point. A companion study (Souza and Costa, 2026) developed a four-equation two-fluid model with dynamic flow-pattern identification that reproduces the cycle in a well–flowline–riser system of field dimensions, located the onset of slugging as the choke opens and showed, with a gas-lift branch added to the system, that injection at 100 to 120 bar regularises the cycle and from 130 bar suppresses it.

This paper is about the operation of that gas-lifted well. Its question is practical: given the two decisions an operator has, the opening of the topside choke and the pressure of the lift gas at the platform, what does the well deliver on average, with what surges and in which regime — and can a model cheap enough for optimisation answer that over the whole plane of the two decisions? The dynamic simulation is too expensive to be called inside an optimiser (minutes per point), so the paper builds a surrogate, and it does so with a method that follows the companion work on compositional steady-state surrogates (Carvalho, Costa, Nunes and Souza, 2026): a multilayer perceptron trained on simulated data, and the same network with a physics term (Raissi, Perdikaris and Karniadakis, 2019; Karniadakis, Kevrekidis, Lu, Perdikaris, Wang and Yang, 2021). The novelty is on both sides of the surrogate. On the data side, the targets are not profiles but time-averaged signatures of a dynamic simulation, and producing them in affordable time required a campaign design of its own: an adaptive horizon that recognises the established regime, warm starts along a shortest path through the sampled points, and a storage that keeps only what the training needs. On the physics side, the governing equations cannot be imposed on a time average of a cycling system; what can, exactly, are the time-averaged mass balances of the well, of the gas-lift branch and of the valve, and the paper shows that this much physics is what the data alone cannot supply.

Section 2 summarises the phenomenological model and the gas-lift branch. Section 3 describes the campaign, the horizon, the markers and the resulting map of the decisions. Section 4 presents the two surrogates and Section 5 compares them. Section 6 formulates the operational optimisation on the surrogate, with capacity constraints on the predicted maxima, and reports its first results.

2. The phenomenological model

The system is the well–flowline–riser of the companion study (Souza and Costa, 2026) with its gas-lift branch (Fig. 1): a vertical well of 1500 m (127 mm) fed by a linear inflow performance relation, a flowline of 2000 m (152.4 mm) whose last 500 m descend at 2° to a low point, a bend of 5 m radius, a riser of 317.5 m and 10 m of horizontal pipe to a choke that discharges into a separator at 10 bar. The gas-lift branch is a 4-in line that runs from the platform along the route of the production line to the wellhead, the casing–tubing annulus (9 5/8 in by 5 1/2 in) to a gas-lift valve 30 m above the bottom of the well, and a check valve fed by a chamber at the base of the annulus. The base well has a reservoir pressure of 200 bar, a productivity index of 5 $\mathrm{m^3\,d^{-1}\,bar^{-1}}$ and a gas–oil ratio of 50 $\mathrm{m^3\,m^{-3}}$; the fluid is the light oil of Carvalho (2022) frozen as a polytropic pair.

Every pipe, the gas pipes included, is described by the one-dimensional four-equation two-fluid model of the companion study: mass and momentum of each phase, closures by flow pattern (stratified, intermittent, bubbly, annular) selected dynamically on the local state through smooth logistic switches, a low-void rule, an objective virtual-mass term and an interfacial pressure in the slug unit that keep the model hyperbolic along the simulated trajectories. The staggered finite-volume discretisation with phase-wise donor cells and the conservative variable-step implicit integration of that study are kept unchanged, with a compiled right-hand side and exact Jacobians; the base mesh has 15 cells in the well, 35 in the flowline–riser pipe, 19 in the gas-lift line and 10 in the annulus, 323 states in all. The gas-lift branch is controlled by one number, the gas pressure P_{gas} at the platform end of the line: the injected rate is not prescribed but follows from P_{gas}, the static gas columns of the line and the annulus, the valve law $q = 0.0075\, C_V \sqrt{\rho_G\, \Delta P}$ ($C_V = 1.5$, ΔP in bar, regularised at the origin) and the pressure of the well at the injection point, so that it responds to whatever the well does; at 20 bar the valve never opens and the system is the base case of the companion study with the branch inert. The reservoir gas is the gas of the well; the lift gas adds to it at the injection point.

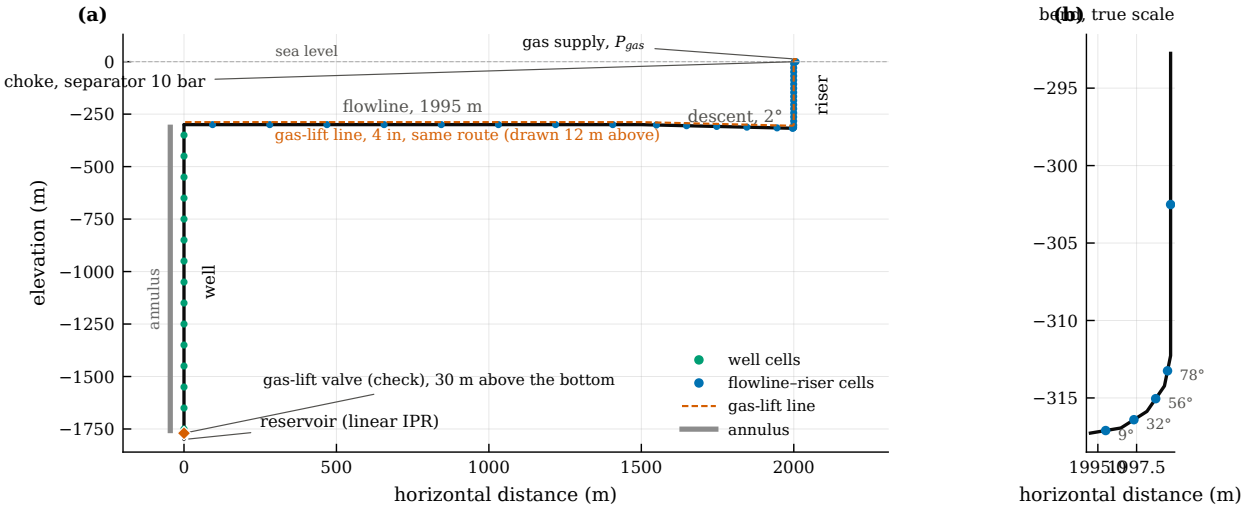


Figure 1: The well–flowline–riser system with the gas-lift branch (schematic, from the companion study): the gas-lift line follows the route of the production line, the annulus runs beside the well and the check valve injects 30 m above the bottom. The two operating decisions are the choke opening a and the platform gas pressure P_{gas}.

3. Data: campaign, horizon and markers

3.1. Sampling the two decisions

The inputs are the choke opening $a \in [0.1, 1]$ and the platform gas pressure P_{gas}, with the well fixed at its base state, in three campaigns: a first over 20 to 160 bar, a second over 160 to 260 bar once the first had placed the optimum of Section 6 on its upper edge, and a third over 160 to 260 bar restricted to openings of 0.5 to 1, the region of that optimum. Points are drawn from a scrambled Sobol' sequence (Sobol', 1967; Owen, 1998) in rounds of 25, so that the rounds of a campaign together remain a single space-filling design, and each point is identified by a hash of its input tuple; only what the training needs is stored, the markers below and the final state of the run. The initial condition of every run is the same: the final state of the base run of the companion study, 8000 s of severe slugging with the choke open and the gas line at 20 bar. The campaigns began with a different rule, one chain per round in which the first point started from the base state and every later point from the final state of the previous one, along the shortest open path through the 25 points on the inputs rescaled to $[-1, 1]$; the rule spares the transient of a cold start, but above an opening of 0.5 with more than 100 bar of lift gas every run started from a neighbour exhausted its budget while every run started from the base state settled in about a minute, and the campaigns were completed from the base state. A steady-state initial condition, the natural alternative, is not attainable for this model: its steady equations are discontinuous in the state, because the closures switch flow pattern at the base of the riser, and a Newton iteration converges only from a state that already has the right pattern in each cell, which a settled dynamic run provides and no algebraic estimate does. The transient from the base state costs 160 to 2000 steps where the well is steady and up to 16000 where it cycles, and it does not depend on the order of the points, so the runs of a round are distributed over processes without any sequencing.

3.2. Adaptive horizon with capture of cycles

Each run integrates in windows of 3000 s and monitors two signals, the pressure at the wellhead and the liquid rate through the choke. After a minimum time (4000 s from a blocked start, 1500 s from a warm start) the regime is taken as established when both signals are permanent (amplitude below 0.1 bar and 0.05 kg s^{-1}) or when two consecutive windows have the same envelope, that is, the same amplitude, mean, minimum and maximum within 10 % of the amplitude; the envelope criterion, rather than equality of periods, is what accepts the cycles whose blowouts alternate in size. When a monitored signal cycles with a period that fits ten times in the window, the comparison window shrinks to ten periods (at least 600 s) and is evaluated on the series already integrated, so that a fast cycle is accepted about 1200 s after it appears instead of two full windows later. The run then continues for three more cycles of the longest monitored period (600 s for a steady state), which are the capture window of the markers, and stops; a run that never

settles stops at 24 000 s and is kept with the markers of its last 4000 s and a flag. A wall-clock budget of 40 min per run in the first rounds, raised to 60 min for the later rounds and for the re-runs of the points that had exhausted it, bounds the points where the integrator crawls: with the choke more than half open and more than 100 bar of lift gas the liquid outflow at the top of the riser oscillates with a period under a minute and an amplitude of up to 15 kg s^{-1}, and the steps fall to 0.3 to 0.6 s with a quarter of them rejected; the short horizontal section between the riser and the choke, the open point of the closures that the companion study reported, is not the cause, since the same runs without it cost the same. Those runs are flagged and excluded from the training.

3.3. Markers at characteristic points

A run is reduced to markers of its capture window at eight points of the network: the bottom of the well, the injection point, the wellhead, the low point of the flowline, the top of the riser, the two ends of the gas-lift line and the base of the annulus. At each point four signals are recorded — pressure, void fraction, liquid and gas mass flow rates — and each signal is summarised by its mean, minimum, maximum and two frequencies, the inverse of the longest and of the shortest interval between successive maxima of the smoothed signal: equal for a periodic cycle, different when blowouts alternate, zero for a permanent signal. The gas injected through the valve is recorded in the same way, and a class is assigned from the wellhead pressure and the choke liquid rate as in the companion study (steady, pressure cycle or severe slugging when the peak liquid rate exceeds twice its mean), with a fourth class, the fast cycle, when the pressure cycles with a period under 300 s: it is the hydrodynamic intermittency of the riser top of Section 3.2, which the peak criterion alone would read as severe slugging although the wellhead pressure swings by a fraction of a bar; the settled severe-slugging and pressure cycles of the campaigns all have periods above 540 s. Trajectories are not kept: they would cost 12 MB per run against 10 kB of markers, and the surrogate consumes only the markers.

3.4. The map of the decisions

Table 1 summarises the three campaigns and Fig. 2 shows them on the plane of the decisions. Of the 291 runs that settled, 99 produce in severe slugging, 13 in a pressure cycle, 3 in the fast cycle and 176 in a steady state; 152 settled on a repeated cycle, 137 on a steady state and 2 reached the time limit, and 29 further runs exhausted the budget (below). The steady class covers two regimes: below 160 bar the well is steady in the strict sense, while above it, with the choke more than half open, 38 of the runs accepted as cycles are the fast cycle of the riser top, with periods of 60 to 90 s and peaks of 7 to 16 kg s^{-1} at means of 5 to 6, which the class rule reads as steady because the wellhead pressure swings by 0.06 to 0.3 bar, less than one per cent of its mean; the three runs classed as fast swing it by 1.5 to 2.2 bar. The stability boundary is a diagonal that runs from about 90 bar of lift gas with the choke at 0.3 to 100 to 125 bar with it open: the more open the choke, the more gas is needed to suppress the cycle, and at the smallest openings some throttled points are steady without lift gas. The check valve opens between 86 and 94 bar almost independently of the choke, so that below 90 bar the data are the base well with the choke as the only decision; between the opening of the valve and the boundary the injection is continuous and modulated by the cycle, which shortens and regularises; above the boundary the well is steady and its production rises with the gas, to 5.3 kg s^{-1} at 160 bar with the choke open, 34 % above the base case, and on to 5.6 kg s^{-1} at 250 bar with the choke half open, a tenth of the gain per bar of the first hundred bar above the boundary. The peak liquid rate through the choke reaches 45 kg s^{-1} in the slugging region with the choke open and stays below 6 kg s^{-1} where the well is steady in the strict sense, rising to 7 to 16 kg s^{-1} on the fast cycle of the corner. The 29 runs that exhausted the budget sit at openings from 0.61 to 0.99: four on the stability boundary of the first campaign, between 110 and 120 bar with the choke from 0.74 to 0.91, in severe slugging, and 25 between 163 and 256 bar, where the riser receives the most gas and the liquid outflow cycles fast (Section 3.2). Twenty-three runs had exhausted the first budget of 40 min and were re-run from the base state with 60: seven settled in a steady state within three minutes (their first failure was the warm start of the early campaigns), four settled on cycles after 37 to 57 minutes and twelve exhausted the hour as well; the other 17 exhausted the hour at their first run, in the rounds completed with the longer budget. All 29 advanced at 0.03 to 0.26 s of wall-clock time per accepted step, the slowest taking 112 000 steps in the hour, and the markers of their partial trajectories already give their class, steady or fast for 24 of them. None enters the training, and they mark where the simulator, not the well, is least efficient.

Table 1
The sampling campaign over the two decisions.

points (runs)	320
steady / fast cycle / pressure cycle / severe slugging	197 / 6 / 14 / 103
stop: steady / cycle captured / time limit / budget	137 / 152 / 2 / 29
simulated time per run (s): median, min–max	9600, 155–24000
accepted steps per run: median, min–max	7027, 159–111975
integration time per run (min): median, max	4.9, 60.0

Base well of Part I ($P_{res} = 200\,\mathrm{bar}$, $J = 5\,\mathrm{m^3\,d^{-1}\,bar^{-1}}$, GOR 50) with the gas-lift branch; inputs sampled by scrambled Sobol' sequences in rounds of 25 over $a \in [0.1, 1]$ and $P_{gas} \in [20, 160]$ bar (first campaign), [160, 260] bar (second) and [160, 260] bar with $a \in [0.5, 1]$ (third, densifying the region of the optimum); every run starts from the final state of the base run (the first rounds chained warm starts along the shortest path, Section 3.1). Budget: 40 min of integration per run in the first rounds, 60 min afterwards and for the re-runs of the points that had exhausted it.

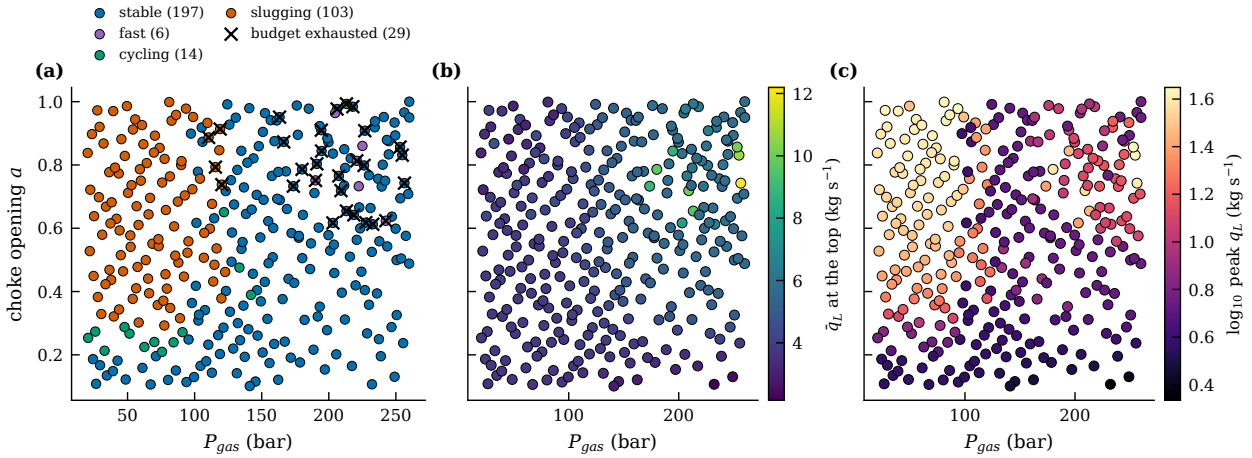


Figure 2: The 320 runs on the plane of the choke opening and the platform gas pressure (the three campaigns and the verification runs of Section 6 that settled): (a) class of the established regime, with the 29 runs that exhausted the integration budget before settling (crosses; their markers come from a partial trajectory and they are excluded from the training); (b) mean liquid rate through the choke; (c) peak liquid rate through the choke on a logarithmic scale.

4. Surrogates

4.1. Outputs and network

The surrogate maps the two decisions to eleven quantities of the capture window: the mean liquid and gas rates at the top of the riser, the mean gas-lift rate through the valve, the spans (maximum minus minimum) of the same three rates, which are the signature of slugging, the maxima of the liquid rate at the top and of the gas-lift rate, which the capacity constraints of the optimisation use, and three auxiliary mean pressures — at the bottom of the well, at the injection point and at the base of the annulus — that the physics of the next section needs. Spans and maxima are learnt as $\log(1 + x)$, so that they are never negative and their orders of magnitude are resolved. The network is a multilayer perceptron (Hornik, Stinchcombe and White, 1989) with two hidden layers of 32 hyperbolic-tangent units and a linear output, inputs and outputs standardised with the training statistics, trained with Adam (Kingma and Ba, 2015) in full batch for 4000 epochs on the mean squared error of the standardised outputs, with the epochs unrolled in a compiled scan so that a network trains in a few seconds; 70 % of the 291 settled runs train it and 30 % test it. The training length matters, and differently for the two networks (Fig. 5d): the total test loss of the data-only network reaches its minimum at about 400 epochs and rises afterwards to 2.4 times that minimum while the training loss keeps falling, and that of the physics-informed network reaches its minimum at about 700 epochs and rises to 1.5 times; the quantities that matter follow. The data-only network overfits them: its test R^2 for the mean liquid rate falls from 0.94 at 1000 epochs to 0.90 at 4000 and 0.86 at 8000, and that of the maximum liquid rate from 0.79 to 0.68 and then below

zero; the physics-informed network holds its means (0.95 to 0.96 over the same range) and loses only slowly on the maxima (0.84 to 0.80): with 204 runs and 1200 weights the physics term is the regulariser of the means. The results below use 4000 epochs for both, so that the comparison shows this difference rather than hiding it behind an early stop. A grid of one or two layers, 8 to 64 units and three learning rates (24 networks of each kind, Section 5) checked that the architecture is not critical.

4.2. Physics of the averages

A physics-informed network adds to the data loss the residual of the governing equations at collocation points that need not be simulated (Raissi et al., 2019). Here the targets are time averages of a dynamic system, and the time average of a cycle does not satisfy the two-fluid equations, which are nonlinear; what it satisfies exactly are the balances that are linear in the averaged quantities. Over a periodic cycle or a steady state, the mass of each phase in the system does not change on average, so that

$$R_1 = \bar{q}_L - J_L\,(P_{res} - \bar{P}_{bh}), \tag{1}$$

$$R_2 = \bar{q}_G - J_G\,(P_{res} - \bar{P}_{bh}) - \bar{q}_{gl}, \tag{2}$$

$$R_3 = \bar{P}_{an} - \kappa\, P_{gas}, \tag{3}$$

$$R_4 = \bar{q}_{gl} - 0.0075\, C_V \sqrt{\rho_G(\bar{P})\,\langle \bar{P}_{an} - \bar{P}_{inj} \rangle_+}, \tag{4}$$

where R_1 and R_2 are the liquid and gas balances between the linear inflow relation at the mean bottom-hole pressure and the mean rates at the top, R_3 is the static gas column from the platform to the base of the annulus ($\kappa = 1.138$ for this fluid and depth), and R_4 is the valve law written with the mean pressures, the only approximate relation of the four (the mean of a square root is not the square root of the mean, but the modulation of the valve pressure drop is within 25 % and the error is a few per cent), with a smooth gate that closes the valve when the mean annulus pressure falls below the mean injection pressure. The maxima, which the capacity constraints use, have no balance of their own, but they have a consistency with the means and the spans that the network can be held to:

$$R_5 = \langle \bar{q}_L - q_L^{max} \rangle_+, \qquad R_6 = \langle \bar{q}_{gl} - q_{gl}^{max} \rangle_+, \tag{5}$$

$$R_7 = \langle q_L^{max} - \bar{q}_L - \Delta q_L \rangle_+, \qquad R_8 = \langle q_{gl}^{max} - \bar{q}_{gl} - \Delta q_{gl} \rangle_+, \tag{6}$$

hinge penalties that are zero when a maximum lies between the mean and the mean plus the span and grow linearly outside, evaluated in physical units whatever the transformation of the targets. The loss is the data term plus λ times the mean square of the eight residuals, each scaled by the standard deviation of the quantity it compares, evaluated on 200 Latin-hypercube points (McKay, Beckman and Conover, 1979) of the plane of the decisions; $\lambda = 1$, and values from 0.1 to 100 give the same fit with residuals decreasing with λ. The residuals are closed-form functions of the network outputs, so that the gradient of the physics term is exact and cheap (the hinges have a kink at zero, which Adam tolerates), unlike the detached-Jacobian construction that the compositional surrogates needed (Carvalho et al., 2026). The information in the physics is therefore not the dynamics but the accounting: the network is told that whatever enters the well on average leaves it on average, and that the lift gas is set by two mean pressures it also predicts.

5. Results

Table 2 and Fig. 3 compare the two networks on the 87 test runs. Both networks predict the mean gas rate at the top and the mean gas-lift rate with R^2 of 0.997 to 0.998 and the three mean pressures with 0.99 to 1.00; the mean liquid rate, 0.90 for the data-only network and 0.96 for the physics-informed one (root-mean-square errors of 0.27 and 0.18 kg s^{-1}), is the hardest of the means: it varies little over the plane (2.2 to 6.1 kg s^{-1}), its time average over three cycles of a surging signal is the noisiest of the targets, and the third campaign, dense in the steady corner, adds points where it changes by a few per cent across the plane. With the first campaign alone (105 training runs) the data-only network reached only 0.57 on it and the physics raised it to 0.88; with the three campaigns the physics add five points of R^2 on it at 4000 epochs, and they show in the residuals: the scaled residual of R_1 (liquid inflow) falls from 0.20 to 0.06 standard units, R_2 to R_4, which both networks satisfy to within 0.08, do not separate them, and the inequalities of the maxima, which the data-only network violates by 0.46 standard units on R_5 and 1.5 on R_7 (maxima above the mean plus the span), fall to 0.10 and 0.28. The maxima are where the physics pay: the maximum lift-gas rate reaches

Table 2
Test metrics of the two surrogates.

output	NN R^2	NN RMSE	PINN R^2	PINN RMSE
$\bar{q}_L$ at the top	0.903	0.272	0.957	0.161
$\bar{q}_G$ at the top	0.998	0.031	0.997	0.040
$\bar{q}_{gl}$ (gas lift)	0.998	0.029	0.997	0.035
span of q_L	-2.598	26.736	0.757	0.895
span of q_G	-0.465	0.895	0.114	1.000
span of q_{gl}	0.165	0.056	0.081	0.059
max. q_L at the top	0.681	6.913	0.818	5.000
max. q_{gl}	0.997	0.041	0.998	0.031
$\bar{P}_{bh}$	0.989	2.172	0.991	1.900
$\bar{P}_{inj}$	0.989	2.171	0.994	1.600
$\bar{P}_{an}$	0.996	4.902	0.999	2.400
residuals R_1–R_8 (RMS, std units)	–	0.203, 0.006, 0.058, 0.078, 0.460, 0.013, 1.537, 0.013	–	0.056, 0.030, 0.024, 0.040, 0.000, 0.000, 0.000, 0.000

Test set of 87 runs (204 for training); RMSE in kg/s for the flow rates and bar for the pressures. Residuals: root-mean-square of the scaled physics residuals (the four balances and the four inequalities of the maxima) on 400 held-out collocation points. Both networks have two hidden layers of 32 hyperbolic-tangent units; the PINN adds the physics term with $\lambda = 1$ on 200 Latin-hypercube collocation points.

0.997 with both networks, since the injection is at most modulated by a quarter, and the maximum liquid rate, which carries the surge, is 0.68 for the data-only network against 0.82 for the physics-informed one at 4000 epochs (0.79 and 0.84 at 1000); the 41 settled runs of the fast cycle, with peaks of 7 to 16 kg s^{-1} at means of 5 to 6 that vary from point to point across the strip (Section 6.2), are the hardest of the maxima. The spans follow: the liquid span of the data-only network fails outright at this training length and the physics-informed one scores 0.76; the spans of the gas and of the injected gas, 0.11 and 0.08, stay beyond both, since the amplitude of a surge in three cycles is a noisy marker by nature and the injection is modulated by a fraction of its mean. Fig. 4 shows the physics-informed surrogate on the plane of the decisions: the mean liquid rate rises with the lift gas above the boundary and falls as the choke closes, the lift-gas rate is zero below 90 bar and rises with P_{gas} as the valve law dictates, and the maximum liquid rate marks the slugging region, with surges of 20 to 40 kg s^{-1} below the boundary against 5 above it and 10 in the corner of the fast cycle, and a narrow band of artefacts along the boundary, where the data change class within a few bar.

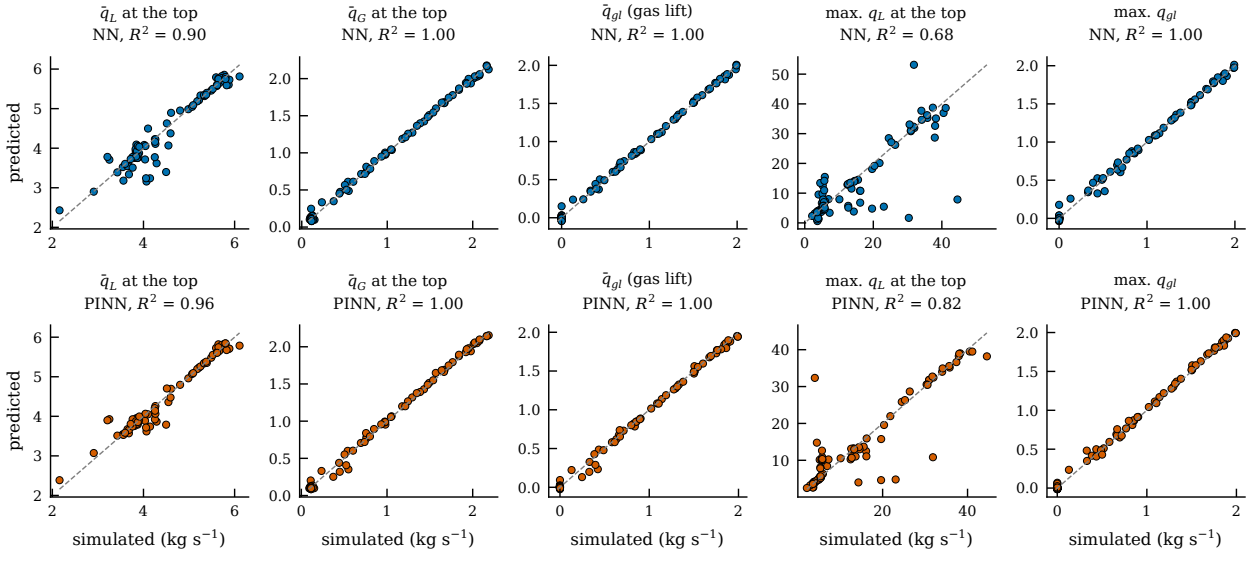


Figure 3: Test parity of the mean liquid, gas and lift-gas rates and of the maxima of the liquid and lift-gas rates, for the data-only network (top row) and the physics-informed network (bottom row).

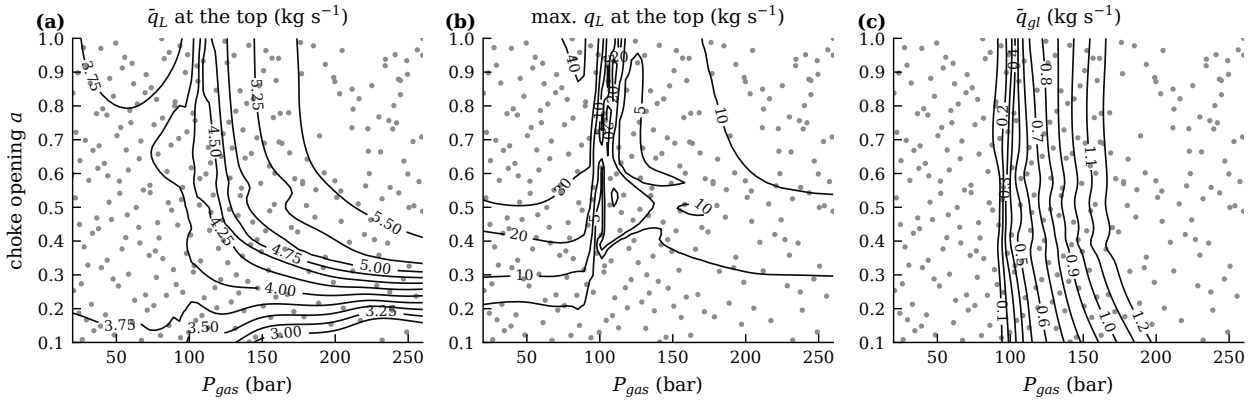


Figure 4: The physics-informed surrogate on the plane of the decisions as labelled level curves, with the positions of the simulated runs (grey dots): (a) mean liquid rate at the top; (b) maximum liquid rate at the top, the surge that reaches the separator; (c) mean gas-lift rate.

Table 3
Sparse-data study: test error of the mean rates against the number of training runs.

training points	$\bar{q}_L$ NN	$\bar{q}_L$ PINN	$\bar{q}_G$ NN	$\bar{q}_G$ PINN	$\bar{q}_{gl}$ NN	$\bar{q}_{gl}$ PINN
10	0.480 ± 0.052	0.518 ± 0.034	0.147 ± 0.032	0.073 ± 0.012	0.150 ± 0.028	0.062 ± 0.011
20	0.344 ± 0.111	0.310 ± 0.091	0.080 ± 0.024	0.055 ± 0.010	0.074 ± 0.016	0.049 ± 0.008
40	0.430 ± 0.182	0.269 ± 0.037	0.076 ± 0.009	0.058 ± 0.002	0.072 ± 0.006	0.052 ± 0.001
204	0.272	0.181	0.031	0.042	0.029	0.039

Test RMSE (kg/s), mean ± standard deviation over three random subsets of the training set; last row: the full training set.

With few data the physics counts more. Table 3 and Fig. 5 retrain both networks on 10, 20 and 40 runs drawn from the training set, three draws each, and test them on the same 87 runs. The physics-informed network has the lower error for the gas and lift-gas rates at every size — half the error of the data-only network with 10 runs, two thirds with 20 and three quarters with 40 — and for the liquid rate from 20 runs on, 0.31 against 0.34 and 0.27 against 0.43 kg s^{-1} with 40, since with ten runs the eight residuals ask more than the data can settle (0.52 against 0.48); with 40 runs its gas error is within 40 % of what it reaches with 204 (0.058 against 0.042 kg s^{-1}) and its liquid error 0.27 against 0.18. The hyperparameter grid confirms that the architecture is not what limits the fit: the mean R^2 over the six requested outputs, the three mean rates and their spans, lies between 0.7 and 0.81 for 12 of the 24 data-only and 15 of the 24 physics-informed architectures and below for the rest, with the spans as the limiting outputs, and the differences between architectures are smaller than the difference the physics makes with few data.

6. Operational optimisation

6.1. Formulation

The operator chooses the choke opening a and the platform gas pressure P_{gas} to maximise the daily margin of the well,

$$F(a, P_{gas}) = c_{oil}\, Q_o(\bar{q}_L) - c_e\, 24\, W(\bar{q}_{gl}, P_{gas}), \tag{7}$$

the revenue of the oil minus the cost of compressing the lift gas, subject to the capacities of the compressor and of the separator,

$$\max q_{gl} \le Q_{gl}^{cap}, \qquad \max q_L \le Q_L^{cap}, \tag{8}$$

written on the maxima of the rates over the cycle, which the surrogate predicts (Section 4.1), and not on their means: a compressor and a separator are sized for peaks. The oil rate Q_o follows from the mean liquid rate with the liquid mass per stock-tank cubic metre of the base fluid (700 kg m^{-3}, oil and solution gas); the compression power W is that of

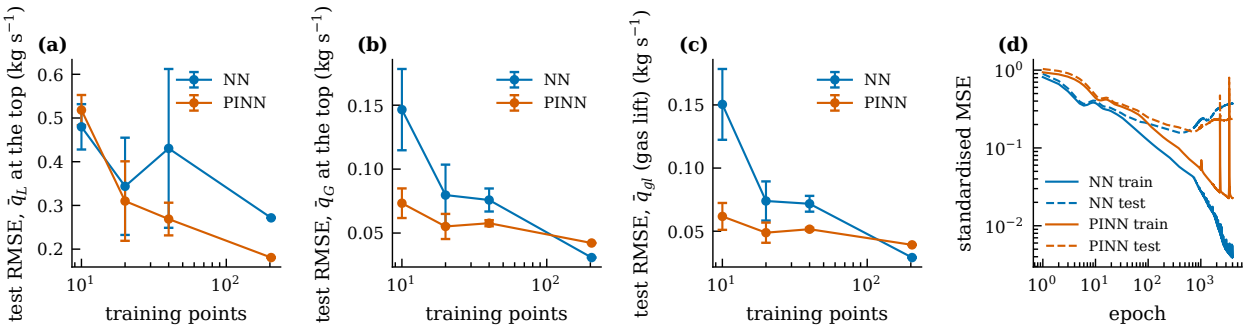


Figure 5: Learning curves of the two surrogates: test root-mean-square error of (a) the mean liquid rate, (b) the mean gas rate and (c) the mean gas-lift rate against the number of training runs (mean and spread of three draws from the training set, and the full training set), evaluated on the same test runs; (d) training loss (solid) and test loss (dashed) of the standardised outputs against the epoch for the two networks, from which the training length is chosen.

a polytropic compression of the mean lift-gas rate from the suction pressure (5 bar) to P_{gas} in stages of ratio at most four with intercooling to the suction temperature ($k = 1.3$, 313 K, efficiency 0.75, molar mass 19.5 g mol^{-1} from the gas density of the fluid). All the quantities in Eqs. (7) and (8) come from the physics-informed surrogate, so that F and the constraints are smooth closed-form functions of the two decisions, evaluated in microseconds. The problem is solved by enumeration on a 201×201 grid of the plane restricted to the feasible nodes, followed by a sequential quadratic programming refinement with the two constraints and the bounds, and the optimum is confirmed by one run of the simulator at that point.

6.2. First results

The example takes a compressor of 1.75 kg s^{-1} as the capacity constraint, an oil price of 70 \$ per barrel and an energy price of 0.10 \$ per kW h. Fig. 6 shows the objective on the plane of the decisions with the runs of the campaigns by class, the level curves of the two predicted peaks and the optimum: $(a, P_{gas}) = (0.82, 227\,\text{bar})$, with a margin of 310×10^3 \$/d, a revenue of 314×10^3 \$/d against a compression cost of 3.2×10^3 \$/d (1350 kW for a mean lift-gas rate of 1.71 kg s^{-1}). The compressor is the active constraint; a separator that accepts peaks of 20 kg s^{-1}, the other constraint of Eq. (8), would not bind, since the predicted peak liquid rate there is 11.6 kg s^{-1}, the fast cycle of the corner. The compression is one per cent of the revenue, so the objective is the production, which rises with the lift gas along the valve law until the compressor stops it; along the constraint the objective is flat to 0.15 % between openings of 0.8 and 1 and to 0.7 % from 0.7, so the opening is not a decision of consequence there (successive fits of the surrogate place the optimum anywhere between 0.8 and 1), and the open choke is the physical bound, not the edge of the sampled box. The simulator run at (1.0, 224 bar), on the constraint 3 bar below the optimum and at the open choke, started from the base state, reproduces the surrogate in the means: over its last 4000 s the liquid rate is 5.79 against the predicted 5.77 kg s^{-1}, the gas rate 1.88 against 1.92, the lift-gas rate 1.71 against 1.72, the bottom-hole pressure 58.4 against 58.0 bar and the annulus pressure 254 against 258. It reproduces the maximum only in its order of magnitude: the run sits on the fast cycle of the riser top described in Section 3.2, with the liquid rate through the choke swinging between 1 and 16 kg s^{-1} every minute while the wellhead pressure stays within 0.3 bar, and the surrogate, trained with the 41 settled runs of that cycle, predicts a peak of 10, the typical amplitude of the strip. The optimum therefore sits in the strip of budget-limited runs of Section 3.4 for a physical reason, the fast cycle, not for a numerical one, and the objective, which depends on the means, is confirmed to 2 %; a separator constraint written on the peak sees the fast cycle, which the settled runs of the strip put in the training data at a cost of 40 to 60 min each, but only at the resolution of its typical amplitude, since that amplitude varies from point to point. A second run on the plateau, at (0.94, 205 bar), agrees with the surrogate in the means and pressures to better than 2 %, and there the fast cycle has an amplitude of only 0.7 kg s^{-1}, so its peak, 5.9, is well below the predicted 10; a third run at (0.95, 260 bar), the far corner of the box, agrees to 1.5 % in every mean and pressure and is steady, with a peak of 5.8 against the same predicted 10: the amplitude of the fast cycle varies across the strip, from nothing to 16 kg s^{-1}, and the markers of three cycles cannot tell the surrogate where. At 260 bar with the choke open the same well is steady, and stays steady when its run is restarted from the settled state. The optimum is insensitive to the oil price, since the compression stays one per cent of the revenue (at 20 \$ per barrel

it does not move), and leaves the constraint only through the energy price: at 2 \$ per kW h the compression is a sixth of the revenue and the optimum retreats to 184 bar with the choke open, and at 5 \$ per kW h it jumps below the opening of the valve, (0.64, 87 bar), the best point of the well without lift gas, since the margin of the lift gas no longer pays its compression.

The constraints change the picture as soon as they bind, and the compressor binds first. The valve law makes the lift-gas rate a function of P_{gas} alone, so a compressor capacity fixes the platform pressure and the optimum walks down the valve law as the capacity shrinks, with a separator that accepts peaks of $20\,\mathrm{kg\,s^{-1}}$ never active: with $1.5\,\mathrm{kg\,s^{-1}}$ the optimum is (1.0, 197 bar) with a margin of 305×10^3 \$/d, two per cent below, since the objective is flat along the valve law at that level; with $1.0\,\mathrm{kg\,s^{-1}}$ it is 140 bar and 282×10^3 \$/d, 9 % below; with $0.8\,\mathrm{kg\,s^{-1}}$ it is (0.85, 121 bar) and 260×10^3 \$/d, 16 % below and just above the stability boundary, with a predicted peak liquid rate of $4.8\,\mathrm{kg\,s^{-1}}$. Along the constraint the objective is nearly indifferent to the opening above 0.7 (it loses 0.1 to 0.6 per cent at 0.9 and four to seven per cent at 0.5, and successive fits of the surrogate place the optimum anywhere between 0.7 and 1), so the choke stays open, except with the smallest of the three compressors, where the open choke at 121 bar is already on the boundary and costs four per cent. A smaller compressor pushes the optimum across the boundary: with $0.5\,\mathrm{kg\,s^{-1}}$ it goes to (0.69, 102 bar), in the slugging region, with surges of $20\,\mathrm{kg\,s^{-1}}$ that a separator of 20 just accepts (it excludes the open choke there); with a separator of $6\,\mathrm{kg\,s^{-1}}$ the separator constraint takes over and holds the optimum at (0.59, 99 bar), at the edge of the cycle, which is what a constraint written on the peak, not on the mean, is for.

The platform gas pressure is often not free: it is set by the compressor and shared with other wells. The operating curve of Fig. 6b and c answers the question the operator then asks, which opening for a given pressure: with P_{gas} fixed, Eq. (7) is a one-dimensional problem in the opening, solved by enumeration on 201 openings and a bounded refinement for each of 121 pressures, under the compressor constraint of the example. Below the opening of the valve the curve runs through the slugging region, with the choke at 0.4 to 0.66 and a margin of 212 to 221×10^3 \$/d, where a separator constraint would move it; at the opening of the valve the choke closes to 0.6 to keep the well out of the cycle, and from 110 bar on it stays open and the margin climbs the valve law, from 229×10^3 \$/d at 100 bar to 262×10^3 at 120, 290×10^3 at 160 and 310×10^3 at 226. Beyond the compressor limit the curve does not end, it turns: a throttled well injects less, since the valve sees a higher tubing pressure, so that the constraint can still be met by closing the choke, to 0.76 at 228 bar, 0.30 at 230 and 0.13 at 260, at a cost of 17 % of the margin at 230 bar and of more than half at 260. The curve thus gives the operator the opening and the margin for any pressure the compressor delivers, and shows that the pressure the example found as optimal is also the pressure beyond which the compressor, not the well, dictates the operation.

6.3. Computational cost

Table 4 puts the cost of the surrogate against that of the simulator it stands for. A run of the campaign compiles the model once per process, in 16 to 27 s, and the compiled code serves the later runs of that process; the integration itself takes from 3 s to the budget of an hour, 2 min on average where the well is steady and 16 where it cycles, since the adaptive horizon stops a steady run as soon as two windows agree and a cycling run only after the capture of three cycles, and the 29 runs that exhausted the budget account for 29 of the 75 process-hours of the three campaigns, which took about three days with five processes in parallel. The physics-informed network trains in 1.3 s for 4000 epochs (the epochs are unrolled in a compiled scan, so that doubling the training length doubles that figure, and the whole study of Section 5, with its grid of 48 networks and the sparse-data draws, takes a few minutes) and is evaluated in 20 µs per point, or 9 ms for the 40 401 points of the grid in one batch, so that the optimisation of Section 6.2, 40 468 evaluations, and the operating curve, 25 717, together take 0.16 s. The same 66 185 evaluations on the simulator, at the mean integration time of a settled run, would take 1.2 process-years. A direct derivative-free search on the simulator, with 200 runs, would cost 32 process-hours, less than half the campaign, and deliver one optimum for one set of prices and capacities; the surrogate delivered the example, the sweep of six capacities, three price sensitivities and the operating curve in less than two seconds, and keeps answering. The campaign is therefore the whole cost of the method, and its size, not the training, is what the design of Section 3.1 is meant to contain: the base-state start that avoids the warm starts that exhausted the budget, the adaptive horizon that stops a steady run in two minutes, and the budget that caps the runs where the integrator crawls.

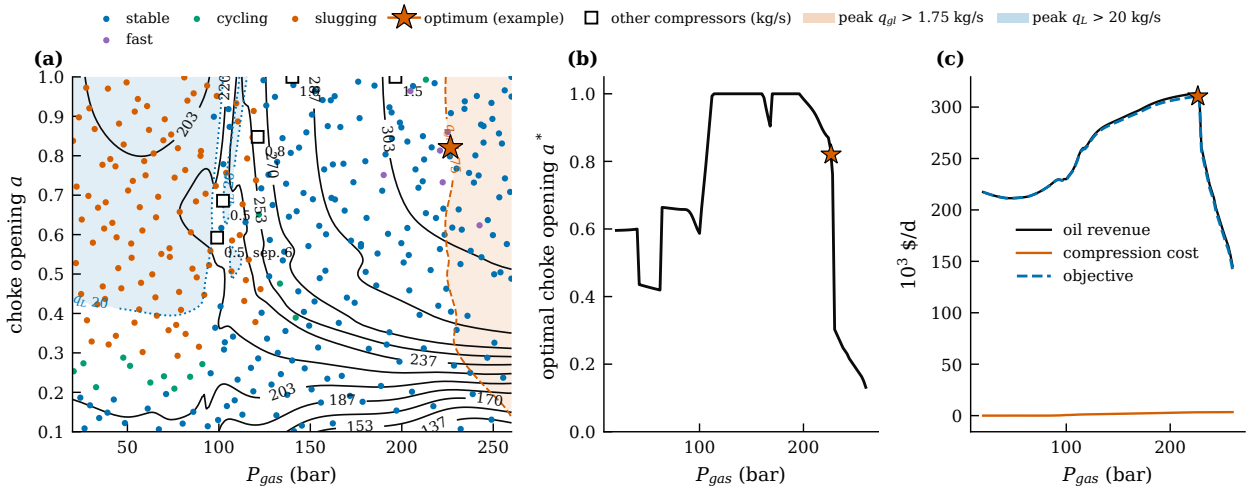


Figure 6: Operational optimisation on the surrogate at 70 $ per barrel and 0.10 $ per kW h: (a) objective of Eq. (7) on the plane of the decisions (level curves, 10^3 $/d), the runs of the campaign by class, the optimum of the example (compressor of $1.75\,\mathrm{kg\,s^{-1}}$; star), the compressor constraint as a level curve and shaded infeasible region (predicted peak lift-gas rate above $1.75\,\mathrm{kg\,s^{-1}}$; dashed, red) and, for reference, the region where the predicted peak liquid rate exceeds $20\,\mathrm{kg\,s^{-1}}$, which a separator of that capacity would exclude (dotted, blue), and the optima for the other compressor capacities of the text (squares, labelled in $\mathrm{kg\,s^{-1}}$; the last one also with a separator of 6); (b) operating curve: the opening that maximises the objective at each platform gas pressure under the compressor constraint, with the optimum of the example (star); (c) revenue, compression cost and objective along that curve.

Table 4
Computational cost of the campaign, of the surrogate and of the optimisation.

compilation of the model (first run of a process; s)	21, 16–27
integration per run, all runs (320; min)	14.1, 0.046–60
integration per run, settled steady (137; min)	2.1, 0.046–28
integration per run, settled on a cycle (152; min)	16.4, 1.5–60
integration per run, budget exhausted (29; min)	60.0, 60–60
campaign, all runs (h)	75
surrogate training, 4000 epochs (s)	1.3 (PINN), 1.7 (NN)
one evaluation of the surrogate (ms)	0.02
grid of 40401 evaluations in one batch (s)	0.009
optimisation, 40468 evaluations (s)	0.13
operating curve, 25717 evaluations (s)	0.03
same evaluations on the simulator, mean settled run (years)	1.2
derivative-free optimisation on the simulator, 200 runs (h)	32

Wall-clock times of the campaigns (JAX backend; each run in one process of the campaign, which XLA spreads over three to four cores of a laptop) and of the surrogate on the same machine, so that the hours are process-hours; the model compiles once per process and the compiled code serves the later runs of that process. The simulator equivalents multiply the number of evaluations of the optimisation and of the operating curve of Section 6.2 by the mean integration time of a settled run; the last row is a direct derivative-free search on the simulator for one set of prices and capacities.

7. Conclusions

A dynamic two-fluid simulation of a gas-lifted well can feed a surrogate of its operating decisions once the campaign is designed for it: one fixed initial condition for every point, the final state of the base run, which the integrator of this model tolerates where a warm start from a neighbour or a Newton steady state does not, an adaptive horizon that recognises the established regime and captures three cycles, and markers at characteristic points instead of trajectories. Over the plane of the choke opening and the platform gas pressure the well has a stability boundary from 90 to 125 bar of lift gas, continuous injection modulated by the cycle below it and 34 % more production above it at 160 bar, 40 % at

250 bar. Of the two surrogates trained on the 291 runs, the physics-informed one, whose physics are the time-averaged mass balances of the well and of the gas-lift branch, predicts the mean liquid rate with R^2 0.96 against 0.90, the gas and lift-gas rates with 0.997 and their maxima with 0.82 against 0.68 and 0.998, holds these figures where the data-only network overfits with the training length, satisfies the well balance and the consistency of the maxima four to five times better, needs half the data for the same error on the gas rates, and leaves the spans of the rates as the quantity to be handled by the class and the stability map rather than by regression. On this surrogate the optimisation of the revenue minus the compression cost, with the predicted maxima bounded by the compressor and separator capacities, is a two-variable problem solved by enumeration and refined in a fraction of a second: with a compressor of 1.75 kg s^{-1}, at the prices of the example, the compression is one per cent of the revenue and the optimum is at about 227 bar with the compressor as the active constraint and the opening indifferent above 0.7; the objective is nearly flat along the valve law, a smaller compressor moves the optimum down the platform pressure to the stability boundary, and a separator capacity written on the peak liquid rate is what keeps it out of the cycle. For a pressure the operator does not control, the operating curve gives the opening and the margin: the choke open from 110 bar to the compressor limit, and throttled beyond it at a steep cost. The campaign is the whole cost: 75 process-hours of simulation against a second of training and a fraction of a second for every optimisation, which on the simulator would take over a year.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used AI-assisted tools for language editing and code assistance. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Data availability

The simulator used in this study is not publicly available; the campaign design (sampling, horizon, markers) and the surrogates are described in full so that they can be reproduced with an independent implementation, and the table of markers and the trained networks are available from the corresponding author upon reasonable request.

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